Annex II. Debt Sustainability Analysis Update

This annex provides an updated Debt Sustainability Analysis (DSA) of Grenada's public and publicly guaranteed (PPG) external and total debt. The macro-framework incorporates all agreed debt restructuring, including the second haircut on commercial debt in 2017 which is contingent on successful completion of the ECF-supported program. With unresolved arrears, including to official bilateral creditors, Grenada's external debt risk rating remains "in debt distress".[1]

1. Grenada's DSA rating remains "in debt distress". In 2016, Grenada started to amortize its restructured commercial bonds while limiting, as agreed under the ECF-supported program, the issuance of new external debt to small amounts of concessional financing from the CDB and World Bank. As a result, the present value of total PPG debt is expected to fall to 61.8 percent of GDP in 2017 from 69.8 percent in 2016, and the present value of public external debt is expected to fall below Grenada's country specific threshold for the first time in years (Figure A2.2). In addition, only one liquidity indicator (debt service to revenue) is breached for only one year (2017) as compared to three instances at the last review.[2] Nevertheless, due to the remaining unresolved arrears to official bilateral creditors, Grenada's DSA rating is unchanged from the December 2016 DSA assessment.

DSA Update: Macroeconomic Assumptions

(In percent of GDP, unless otherwise specified)

	Historical	Projections			
		2017-22		2023-36	
	Average	5th Review	6th Review	5th Review	6th Review
Non-interest current account balance	-18.7	-14.7	-13.1	-14.3	-13.7
Real GDP growth (in percent)	2.7	2.7	2.5	2.7	2.7
Growth of exports of G & S (USD terms, in perce	6.3	5.3	5.0	5.8	5.4
Current official transfer	-3.1	-2.6	-2.1	-2.1	-1.8
Net FDI	-8.3	-9.0	-9.0	-9.1	-9.1
Primary balance	-0.9	2.7	4.0	-1.0	-1.0
Revenue and grants	23.6	24.3	26.5	23.2	24.5
of which: grants	3.0	2.4	4.1	2.2	2.0
Primary (non-interest) expenditure	24.9	21.6	21.8	24.2	25.5
Inflation rate (GDP deflator, in percent)	1.6	2.4	2.2	2.3	2.1
Memorandum item		5th Review	6th Review		
2016 Nominal GDP (in million USD)		1,029.8	1,037.9		

Source: Grenadian authorities and IMF staff projections

[1] See Country Report No. 16/38. The CPIA ranks Grenada as a medium performer, with an average CPIA rating of 3.52 in 2013-2015.

[2] This does not include contingent liabilities related to the potential repurchase of shares in Grenlec. An independent valuation of the shares is required and government may choose non-debt financing options.

2. Sustained strong efforts are crucial for fully restoring the sustainability of Grenada's public debt. The debt restructuring achieved so far, along with stronger economic growth and sizeable fiscal adjustment, has put debt dynamics on a steady downward trajectory. Having completed the debt restructuring with Russia and the United Kingdom in early 2017, the remaining restructurable official debt consists of US$52 million in debt to non-Paris Club official bilateral creditors, for which the authorities are seeking treatment comparable with the agreements with commercial creditors. One external private creditor is seeking payment of its claims through the Grenadian court system. The government is servicing this debt according to: (i) the court judgment on one portion and (ii) the government's proposed repayment terms. On the domestic front, negotiations have been concluded with Petrocaribe Grenada, with two banks, and with the majority of holders of a domestic bond. The restructuring of government guaranteed debt has also been concluded.

3. The prospects for restoring Grenada's debt sustainability are subject to downside risks. In particular, the baseline scenario assumes strict adherence to the Fiscal Responsibility Act (FRA). Moreover, the historical scenario (Figure A2:2) illustrates how a return to lax fiscal policies would put Grenada's public debt quickly back on an unsustainable path. The experience of the past decade, with moderate GDP growth, also highlights the tourism sector's exposure to macroeconomic conditions in advanced economies and to natural disaster risks. A customized shock scenario (Figure A2.I) analyzes the impact of the latter. Further risks to the outlook are posed by shocks to import prices, particularly fuel prices. The authorities reiterated their commitment to the fiscal responsibility law and to concluding the debt restructuring as soon as possible.

Table A2.1. Grenada: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014 – 2037

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	101.8	91.7	83.4			71.8	65.8	60.4	54.9	49.8	46.3		46.3	44.2	
of which: foreign-currency denominated	67.7	62.6	57.9			48.3	44.8	42.2	38.9	34.8	32.6		30.6	26.8	
Change in public sector debt	-6.3	-10.0	-8.3			-11.6	-6.0	-5.4	-5.6	-5.1	-3.5		0.0	-2.4	
Identified debt-creating flows	-3.5	-12.1	-8.7			-11.4	-5.9	-5.3	-5.7	-5.0	-3.4		-0.1	0.8	
Primary deficit	1.1	-2.2	-5.3	0.9	3.1	-4.4	-4.3	-4.3	-4.4	-3.9	-2.4	-4.0	1.0	2.0	1.1
Revenue and grants	24.5	24.8	26.8			26.5	26.1	25.9	25.7	25.3	25.0		24.9	23.0	
of which: grants	4.1	3.3	3.6			4.2	3.8	3.5	3.3	2.9	2.5		2.5	1.5	
Primary (noninterest) expenditure	25.6	22.6	21.5			22.0	21.8	21.6	21.2	21.4	22.6		26.0	25.0	
Automatic debt dynamics	-4.6	-4.1	-1.8			-1.7	-1.1	-1.0	-1.2	-1.2	-1.0		-1.1	-1.2	
Contribution from interest rate/growth differential	-4.6	-4.1	-1.8			-2.0	-1.1	-1.0	-1.3	-1.2	-1.0		-1.0	-1.1	
of which: contribution from average real interest rate	2.8	1.9	1.6			0.0	0.5	0.4	0.2	0.2	0.3		0.3	0.1	
of which: contribution from real GDP growth	-7.4	-6.0	-3.5			-2.1	-1.6	-1.4	-1.5	-1.5	-1.3		-1.2	-1.2	
Contribution from real exchange rate depreciation	0.0	0.0	0.0			0.3	0.1	0.0	0.1	0.1	0.1		...	...	
Other identified debt-creating flows	0.0	-5.8	-1.6			-5.3	-0.5	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)6/	0.0	-5.8	-1.0			-5.3	-0.5	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	-0.6			0.1	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-2.8	2.0	0.4			-0.2	-0.1	-0.1	0.1	-0.1	-0.1		0.1	-3.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	69.8			61.8	56.4	51.3	46.3	42.1	38.8		37.7	37.7	
of which: foreign-currency denominated	...	...	44.2			38.3	35.4	33.0	30.4	27.1	25.2		22.0	20.3	
of which: external	...	...	44.2			38.3	35.4	33.0	30.4	27.1	25.2		22.0	20.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	22.3	17.3	15.8			13.7	11.7	11.2	10.2	10.0	10.9		11.7	9.0	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	260.2			233.7	215.7	198.1	180.5	166.4	155.5		151.2	163.6	
PV of public sector debt-to-revenue ratio (in percent)	...	...	300.2			277.6	251.9	229.1	206.7	187.7	173.2		167.9	175.4	
of which: external 3/	...	...	190.3			172.0	158.0	147.4	135.4	121.0	112.2		98.1	94.4	
Debt service-to-revenue and grants ratio (in percent) 4/	86.6	78.5	78.7			68.6	61.3	59.6	57.0	54.9	53.1		42.7	30.5	
Debt service-to-revenue ratio (in percent) 4/	103.9	90.5	90.7			81.4	71.5	68.9	65.2	62.0	59.1		47.4	32.7	
Primary deficit that stabilizes the debt-to-GDP ratio	7.4	7.9	3.0			7.2	1.7	1.1	1.2	1.2	1.1		1.0	4.3	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.3	6.2	3.9	2.7	3.3	2.5	2.3	2.2	2.5	2.7	2.7	2.5	2.7	2.7	2.7
Average nominal interest rate on forex debt (in percent)	3.0	3.7	3.6	2.7	0.7	3.2	2.9	2.7	2.2	2.5	2.7	2.7	2.1	1.9	2.0
Average real interest rate on domestic debt (in percent)	3.9	1.8	1.4	2.5	1.2	0.1	1.0	1.0	1.1	1.0	1.0	0.9	1.1	0.5	1.1
Real exchange rate depreciation (in percent, + indicates depreciation)	0.0	0.0	0.0	0.0	0.0	0.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.8	1.6	1.5	1.6	1.4	2.6	2.2	2.2	2.1	2.1	2.1	2.2	2.1	2.1	2.1
Growth of real primary spending (deflated by GDP deflator, in percent)	10.8	-6.2	-1.1	0.5	5.1	4.9	1.2	1.4	0.6	3.7	8.3	3.4	2.5	2.0	3.5
Grant element of new external borrowing (in percent)	...	...	...	...	...	39.1	33.3	32.7	33.1	29.7	29.7	32.9	29.7	29.7	29.7

Sources: Country authorities; and staff estimates and projections.

1/ Comprises central government and government guaranteed debt

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

6/Debt relief includes savings from restructuring agreements reached with external and domestic creditors.

Table A2.2. Grenada: External Debt Sustainability Framework, Baseline Scenario, 2014 – 2037[1]

(In percent of GDP, unless otherwise indicated)

	Actual											Historical Average 6/	Standard Deviation 6/	Projections									
	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037	2023-2037 Average
External debt (nominal) 1/	**...**	**...**	**...**	**113.5**	**123.4**	**136.3**	**141.0**	**144.7**	**144.8**	**137.1**	**130.2**			**118.9**	**114.6**	**111.2**	**107.1**	**102.0**	**98.8**		**92.1**	**79.7**	
of which: public and publicly guaranteed (PPG)	...	...	...	63.9	68.9	69.0	67.6	70.8	67.7	62.6	57.9			48.3	44.8	42.2	38.9	34.8	32.6		30.6	26.8	
Change in external debt	...	...	...	...	9.9	12.9	4.6	3.7	0.1	-7.7	-6.9			-11.3	-4.3	-3.3	-4.1	-5.1	-3.2		-1.3	-2.2	
Identified net debt-creating flows	...	...	...	...	15.9	16.9	13.5	2.6	2.5	-3.1	-0.9			3.7	3.7	3.4	2.6	2.4	3.1		3.3	2.6	
Non-interest current account deficit	**...**	**...**	**...**	**22.6**	**22.2**	**22.0**	**19.0**	**21.3**	**15.4**	**15.0**	**12.3**	**18.7**	**4.0**	**13.5**	**13.5**	**13.6**	**13.4**	**13.3**	**13.8**		**14.1**	**12.9**	13.9
Deficit in balance of goods and services	...	...	...	21.3	22.7	22.6	20.6	21.9	16.3	15.2	12.4			14.3	14.1	14.1	13.7	13.8	14.4		14.1	12.3	
Exports	...	...	...	25.5	26.6	28.2	28.9	28.5	27.9	27.2	29.0			28.2	28.4	28.7	29.0	29.1	29.3		29.9	30.0	
Imports	...	...	...	46.8	49.2	50.8	49.5	50.4	44.2	42.4	41.4			42.5	42.5	42.8	42.7	42.9	43.7		44.1	42.3	
Net current transfers (negative = inflow)	...	...	...	-5.1	-4.1	-3.1	-3.8	-2.2	-2.4	-1.8	-2.6	-3.1	1.1	-2.4	-2.2	-2.1	-2.1	-2.2	-2.2	-2.2	-1.9	-1.4	-1.8
of which: official	...	...	...	-3.0	-1.9	-0.9	-1.4	-0.2	-0.7	-0.2	-1.1			-1.0	-0.8	-0.6	-0.6	-0.6	-0.5		-0.4	-0.2	
Other current account flows (negative = net inflow)	...	...	...	6.4	3.6	2.5	2.2	1.7	1.5	1.5	2.5			1.6	1.6	1.6	1.8	1.7	1.6		1.9	2.0	
Net FDI (negative = inflow)	**...**	**...**	**...**	**-13.3**	**-7.8**	**-5.5**	**-3.9**	**-13.4**	**-4.1**	**-9.9**	**-8.3**	**-8.3**	**3.7**	**-8.6**	**-8.7**	**-9.2**	**-9.2**	**-9.2**	**-9.2**	-9.0	**-9.2**	**-8.7**	-9.1
Endogenous debt dynamics 2/	**...**	**...**	**...**	**...**	**1.6**	**0.4**	**-1.5**	**-5.3**	**-8.8**	**-8.2**	**-4.8**			**-1.2**	**-1.1**	**-1.1**	**-1.6**	**-1.7**	**-1.6**		**-1.7**	**-1.6**	
Contribution from nominal interest rate	...	...	...	...	1.5	1.6	2.1	1.9	2.1	2.5	2.3			1.9	1.5	1.3	1.1	1.1	1.1		0.8	0.5	
Contribution from real GDP growth	...	...	...	...	0.6	-0.9	1.5	-3.1	-9.8	-8.4	-5.1			-3.2	-2.6	-2.4	-2.7	-2.8	-2.7		-2.4	-2.1	
Contribution from price and exchange rate changes	...	...	...	...	-0.5	-0.3	-5.2	-4.0	-1.1	-2.3	-2.0			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**...**	**...**	**...**	**...**	**-6.0**	**-3.9**	**-8.9**	**1.1**	**-2.4**	**-4.6**	**-6.0**			**-15.0**	**-8.0**	**-6.7**	**-6.8**	**-7.5**	**-6.2**		**-4.6**	**-4.8**	
of which: exceptional financing	...	...	...	...	0.0	0.0	-0.2	-0.7	-3.4	-3.1	-1.5			-1.2	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
debt restructuring									0.0	-5.8	-4.7			-0.6	-4.3	0.0	0.0	0.0	0.0				
capital transfers														-12.0	-3.7	-7.9	-7.7	-7.3	-7.0		-6.8	0.0	
PV of external debt 4/	...	...	...	...	...	...	...	...	...	...	116.6			108.9	105.2	102.1	98.6	94.4	91.4		83.6	73.2	
In percent of exports	...	...	...	...	...	...	...	...	...	...	402.1			386.8	369.7	355.2	340.4	324.4	312.2		279.3	244.1	
PV of PPG external debt	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**44.2**			**38.3**	**35.4**	**33.0**	**30.4**	**27.1**	**25.2**		**22.0**	**20.3**	
In percent of exports	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**152.6**			**136.1**	**124.3**	**114.9**	**104.8**	**93.3**	**86.0**		**73.7**	**67.6**	
In percent of government revenues	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**190.3**			**172.0**	**158.0**	**147.4**	**135.4**	**121.0**	**112.2**		**98.1**	**94.4**	
Debt service-to-exports ratio (in percent)	**...**	**...**	**...**	**6.8**	**5.7**	**5.6**	**7.2**	**6.5**	**13.3**	**16.8**	**19.1**			**20.5**	**16.1**	**14.3**	**13.5**	**12.3**	**11.7**		**9.3**	**6.2**	
PPG debt service-to-exports ratio (in percent)	**...**	**...**	**...**	**4.9**	**4.3**	**4.4**	**6.4**	**5.9**	**12.6**	**16.1**	**18.5**			**19.8**	**15.4**	**13.7**	**12.9**	**11.7**	**11.1**		**8.9**	**6.2**	
PPG debt service-to-revenue ratio (in percent)	**...**	**...**	**...**	**6.4**	**5.6**	**6.1**	**9.4**	**8.6**	**17.2**	**20.3**	**23.0**			**25.0**	**19.6**	**17.5**	**16.6**	**15.1**	**14.5**		**11.8**	**8.7**	
Total gross financing need (Millions of U.S. dollars)	...	...	...	...	122.4	140.8	137.0	82.4	136.7	94.9	98.4			116.6	107.3	102.5	101.4	100.8	110.9		134.9	170.8	
Non-interest current account deficit that stabilizes debt ratio	...	...	...	...	12.3	9.0	14.3	17.6	15.3	22.7	19.2			24.8	17.8	17.0	17.5	18.3	17.0		15.4	15.1	
Key macroeconomic assumptions																							
Real GDP growth (in percent)	...	...	...	...	-0.5	0.8	-1.2	2.4	7.3	6.2	3.9	2.7	3.3	2.5	2.3	2.2	2.5	2.7	2.7	2.5	2.7	2.7	2.7
GDP deflator in US dollar terms (change in percent)	...	...	...	...	0.5	0.2	3.9	2.9	0.8	1.6	1.5	1.6	1.4	2.6	2.2	2.2	2.1	2.1	2.1	2.2	2.1	2.1	2.1
Effective interest rate (percent) 5/	...	...	...	...	1.3	1.3	1.6	1.4	1.6	1.9	1.8	1.5	0.2	1.6	1.3	1.2	1.0	1.1	1.1	1.2	0.9	0.6	0.8
Growth of exports of G&S (US dollar terms, in percent)	...	...	...	...	3.9	7.3	5.2	3.8	6.1	5.1	12.4	6.3	3.0	2.2	5.6	5.5	5.4	5.4	5.6	5.0	5.3	0.0	5.1
Growth of imports of G&S (US dollar terms, in percent)	...	...	...	...	5.1	4.2	0.0	7.2	-5.0	3.5	2.9	2.6	4.0	8.0	4.6	5.2	4.2	5.4	6.8	5.7	5.1	-0.9	4.7
Grant element of new public sector borrowing (in percent)		...	...	...	...	...	...	...	...	...	...	...	...	39.1	33.3	32.7	33.1	29.7	29.7	32.9	29.7	29.7	29.7
Government revenues (excluding grants, in percent of GDP)	...	...	...	19.5	20.6	20.3	19.7	19.5	20.4	21.5	23.2			22.3	22.4	22.4	22.4	22.4	22.4	22.4	22.5	21.5	22.4
Aid flows (in Millions of US dollars) 7/	...	...	...	25.3	31.1	26.1	8.8	11.6	37.2	32.3	37.0			65.5	50.6	49.6	48.4	37.7	35.0		43.1	43.4	
of which: Grants	...	...	...	25.3	31.1	26.1	8.8	11.6	37.2	32.3	37.0			45.6	42.8	41.7	40.6	37.7	35.0		43.1	43.4	
of which: Concessional loans	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0			19.8	7.8	7.8	7.8	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...	...	...	...	...	...	...	...	...			5.9	4.6	4.2	3.7	2.9	3.1	4.1	3.4	1.9	2.8
Grant-equivalent financing (in percent of external financing) 8/	...	...	...	...	...	...	...	...	...	...	...			68.7	73.0	75.0	79.8	95.0	71.2		60.9	67.5	60.3
Memorandum items:																							
Nominal GDP (Millions of US dollars)	...	...	...	771.3	771.0	778.7	799.9	842.6	911.6	984.1	1037.9			1091.9	1141.5	1192.2	1247.6	1308.6	1372.5		1740.3	2803.7	
Nominal dollar GDP growth	...	...	...	...	0.0	1.0	2.7	5.3	8.2	8.0	5.5			5.2	4.5	4.4	4.6	4.9	4.9	4.8	4.9	4.9	4.9
PV of PPG external debt (in Millions of US dollars)											459.2			418.4	403.7	393.7	378.7	355.2	345.5		383.7	568.4	
(PVt-PVt-1)/GDPt-1 (in percent)														-3.9	-1.3	-0.9	-1.3	-1.9	-0.7	-1.7	0.6	0.1	0.7
Gross workers' remittances (Millions of US dollars)	...	...	...	27.5	28.3	29.0	29.7	30.5	31.1	31.7	32.6			33.9	35.4	39.8	44.3	49.0	53.9		65.5	93.2	
PV of PPG external debt (in percent of GDP + remittances)									...	...	42.9			37.2	34.3	32.0	29.3	26.2	24.2		21.2	19.6	
PV of PPG external debt (in percent of exports + remittances)									...	...	137.7			122.6	112.1	102.9	93.4	82.6	75.8		65.5	60.9	
Debt service of PPG external debt (in percent of exports + remittances)									...	...	16.7			17.8	13.9	12.2	11.4	10.3	9.8		7.9	5.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A2.3. Grenada: Sensitivity Analysis for Key Indicators of Public Debt 2017 – 2037

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	62	56	51	46	42	39	38	38
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	62	60	59	57	57	55	53	48
A2. Primary balance is unchanged from 2017	62	56	51	46	41	37	16	-19
A3. Permanently lower GDP growth 1/	62	57	52	48	45	42	47	67
A4. Alternative Scenario: Natural Disaster	62	61	56	51	48	45	47	52
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	62	59	56	51	48	45	48	56
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	62	63	64	58	54	50	49	46
B3. Combination of B1-B2 using one half standard deviation shocks	62	63	63	58	54	50	51	52
B4. One-time 30 percent real depreciation in 2018	62	73	67	62	57	53	49	47
B5. 10 percent of GDP increase in other debt-creating flows in 2018	62	64	59	53	49	46	44	43
PV of Debt-to-Revenue Ratio 2/								
Baseline	234	216	198	180	166	155	151	164
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	234	231	227	224	224	222	212	209
A2. Primary balance is unchanged from 2017	234	215	197	180	164	147	65	-82
A3. Permanently lower GDP growth 1/	234	218	202	187	175	168	187	290
A4. Alternative Scenario: Natural Disaster	234	248	229	213	199	188	202	231
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	234	223	214	199	188	180	193	242
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	234	241	247	227	212	201	195	200
B3. Combination of B1-B2 using one half standard deviation shocks	234	239	243	224	211	201	203	224
B4. One-time 30 percent real depreciation in 2018	234	279	259	240	225	213	195	203
B5. 10 percent of GDP increase in other debt-creating flows in 2018	234	246	226	208	194	183	178	185
Debt Service-to-Revenue Ratio 2/								
Baseline	69	61	60	57	55	53	43	30
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	69	61	60	57	56	54	45	36
A2. Primary balance is unchanged from 2017	69	61	60	57	55	53	41	18
A3. Permanently lower GDP growth 1/	69	62	60	58	56	55	46	39
A4. Alternative Scenario: Natural Disaster	69	66	64	61	59	56	47	34
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	69	63	63	60	58	56	46	36
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	69	61	61	59	56	54	45	34
B3. Combination of B1-B2 using one half standard deviation shocks	69	62	62	60	57	55	46	36
B4. One-time 30 percent real depreciation in 2018	69	65	66	63	61	59	48	38
B5. 10 percent of GDP increase in other debt-creating flows in 2018	69	61	61	58	56	54	45	33

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table A2.4. Grenada: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt 2017 – 2037

(In percent)

				Projections				
	2017	2018	2019	2020	2021	2022	**2027**	2037
PV of debt-to GDP ratio								
Baseline	38	35	33	30	27	25	**22**	20
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	38	39	41	43	44	47	**63**	78
A2. New public sector loans on less favorable terms in 2017-2037 2	38	36	34	32	29	27	**28**	30
A3. Alternative Scenario: Natural Disaster	38	39	37	34	31	29	**25**	21
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	38	36	35	32	29	27	**23**	21
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	38	36	34	32	28	26	**23**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	38	36	34	32	28	26	**23**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	38	38	39	37	33	31	**27**	21
B5. Combination of B1-B4 using one-half standard deviation shocks	38	38	39	36	32	30	**27**	22
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	38	50	47	43	38	36	**31**	29
PV of debt-to-exports ratio								
Baseline	136	124	115	105	93	86	**74**	68
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	136	139	144	149	153	159	**209**	260
A2. New public sector loans on less favorable terms in 2017-2037 2	136	127	119	110	99	92	**92**	100
A3. Alternative Scenario: Natural Disaster	136	138	128	117	106	98	**83**	69
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	136	124	115	105	93	86	**74**	68
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	136	129	125	114	102	94	**81**	72
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	136	124	115	105	93	86	**74**	68
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	136	135	137	127	115	107	**92**	72
B5. Combination of B1-B4 using one-half standard deviation shocks	136	131	130	120	108	100	**86**	71
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	136	124	115	105	93	86	**74**	68
PV of debt-to-revenue ratio								
Baseline	172	158	147	135	121	112	**98**	94
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	172	176	184	192	198	208	**278**	363
A2. New public sector loans on less favorable terms in 2017-2037 2	172	161	153	142	128	121	**123**	139
A3. Alternative Scenario: Natural Disaster	172	175	164	151	137	128	**111**	97
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	172	163	156	143	128	119	**104**	100
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	172	160	153	141	127	118	**103**	96
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	172	161	153	141	126	116	**102**	98
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	172	172	176	164	149	140	**122**	100
B5. Combination of B1-B4 using one-half standard deviation shocks	172	169	173	160	145	135	**118**	103
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	172	224	209	192	171	159	**139**	134

Table A2.4. Grenada: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017 – 2037 (concluded)

(In percent)

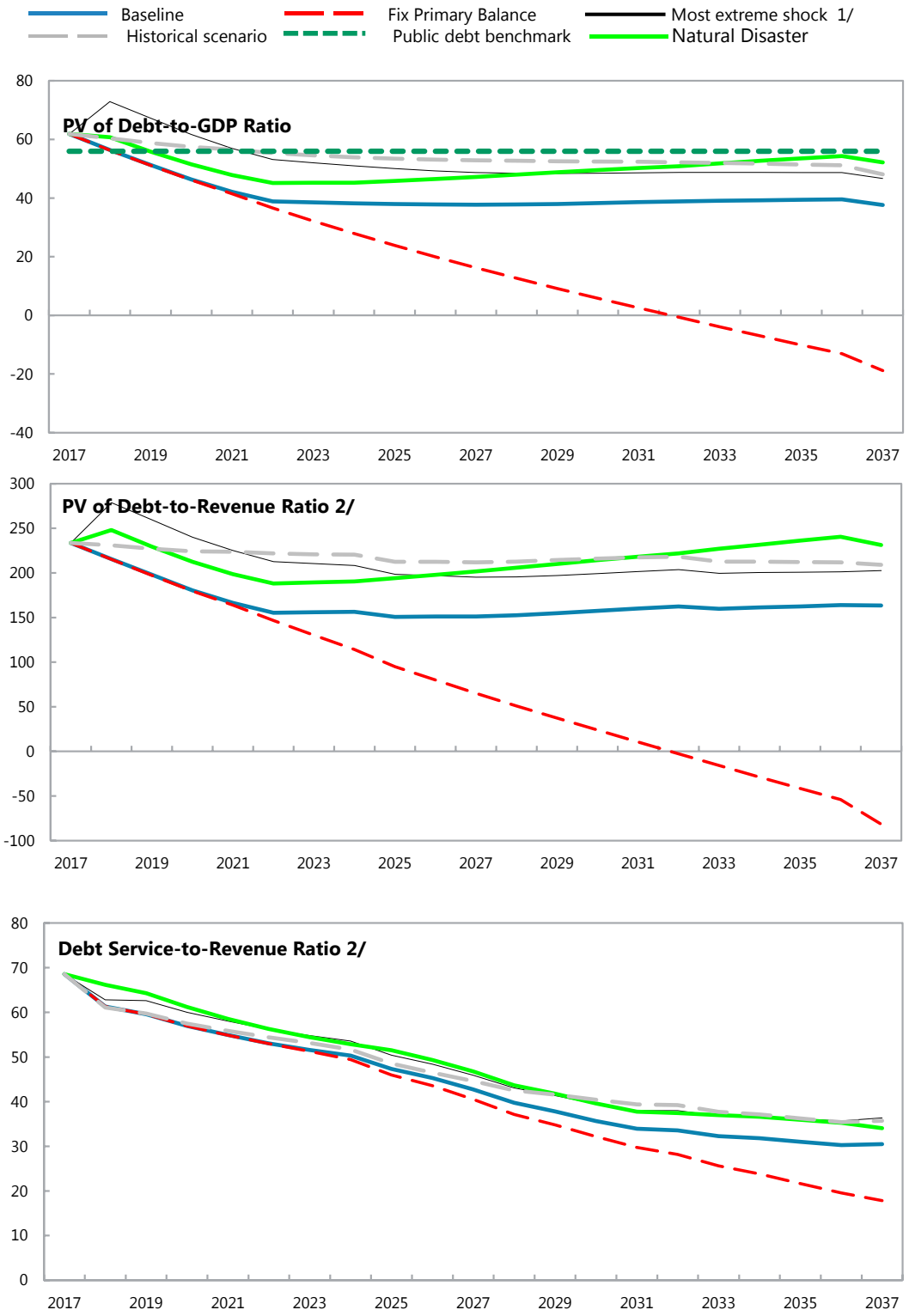
	Projections							
	2017	2018	2019	2020	2021	2022	**2027**	2037
Debt service-to-exports ratio								
Baseline	20	15	14	13	12	11	**9**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	20	15	14	14	13	13	**14**	22
A2. New public sector loans on less favorable terms in 2017-2037 2	20	15	14	13	12	11	**9**	9
A3. Alternative Scenario: Natural Disaster	20	16	14	13	12	11	**10**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	20	15	14	13	12	11	**9**	6
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	20	16	14	14	12	12	**10**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	20	15	14	13	12	11	**9**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	20	15	14	13	12	12	**11**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	20	16	14	13	12	12	**10**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	20	15	14	13	12	11	**9**	6
Debt service-to-revenue ratio								
Baseline	25	20	18	17	15	14	**12**	9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	25	20	18	18	17	17	**19**	31
A2. New public sector loans on less favorable terms in 2017-2037 2	25	20	17	17	15	15	**12**	12
A3. Alternative Scenario: Natural Disaster	25	20	18	17	16	15	**13**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	25	20	19	18	16	15	**12**	9
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	25	20	18	17	15	15	**12**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	25	20	18	17	16	15	**12**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	25	20	18	17	16	15	**14**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	25	20	19	18	16	16	**14**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	25	28	25	23	21	21	**17**	12
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	29	29	29	29	29	29	**29**	29

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

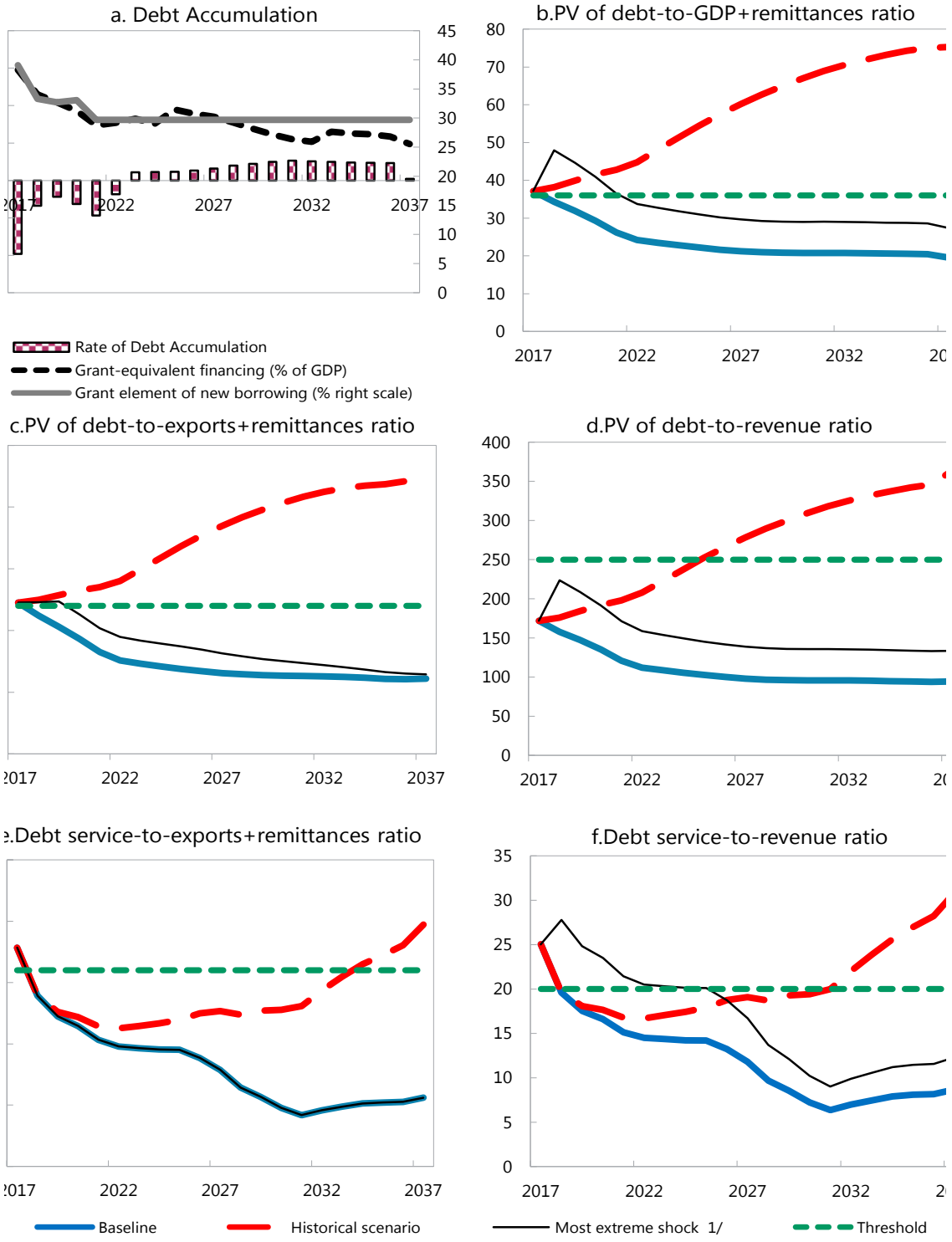
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming and offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure A2.1. Grenada: Indicators of Public Debt Under Alternative Scenarios, 2017 – 2037[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.

Figure A2.2. Grenada: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2017 – 2037[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a One-time depreciation shock; in c. to a Non-debt flows shock; in d. to a One-time depreciation shock; in e. to a Growth shock and in figure f. to a One-time depreciation shock